LIBRARY MANAGEMENT SYSTEM

Python Midterm Project Report

Team members: Add the three names before submission.
Course: Programming Principles 2
Submission: October 18, 2026

Project overview

This is a small console program for managing a library. It stores books, members, and borrowing records in SQLite. A short menu lets the user add, display, search, update, and delete records, then view simple counts or save a text report.

Database

Table	Fields	Purpose
books	book_id, title, author, available	Stores each book and whether it is available
members	member_id, name, email	Stores library members
borrowings	borrowing_id, book_id, member_id, returned	Connects a member with a borrowed book

The book_id and member_id fields in borrowings are foreign keys. The database has three related tables.

How the program works

Classes

Book stores a book ID, title, author, and availability. Its display method prints the details on one line. **Member** stores a member ID, name, and email and prints the member details.

Functions

Function	Action
add_book / add_member	Save a new book or member
borrow_book / delete_record	Create a borrowing or cancel a record
display_records / search_records	Show books, members, borrowing records, or matching results
update_book	Change a book title or author
show_statistics	Count borrowing records and active loans; find the most borrowed book
export_results	Write a summary and borrowing list to results.txt

Python concepts from class

The code uses basic variables, input, output, conditions, and a while loop for the menu. Functions handle each action. Book and Member are simple classes. SQLite stores the data, and a text file saves results. Regular expressions check member email addresses; random selects the sample borrowing records.

Input checks

The program checks that required names are present, validates member email format, asks for number IDs, and prevents borrowing a book that is already out.

Sample data and results

Dataset

The included database has 50 books, 30 members, and 30 borrowings. Ten borrowing records are marked returned; 20 books are still borrowed. The names and records are fictional sample data.

Statistics

Calculation	Sample result
Total borrowing records	30
Books currently borrowed	20
Most borrowed book	The sample has distinct book titles in the loan records; the top count is one.

Output from the running program

```
Choose an option: 9
Total borrowing records: 30
Books currently borrowed: 20
Most borrowed book: A Long Journey 1 - 1 time(s)
```

Team contributions

Team member	Contribution
Member 1 - add name	Add the work completed and presentation part.
Member 2 - add name	Add the work completed and presentation part.
Member 3 - add name	Add the work completed and presentation part.

Run with python3 main.py. Choose option 10 to create or refresh results.txt.